



Gaze-Controlled Virtual Keyboard System

Akhil Saladi & Eraj Wali

Computer Systems Lab
2026 - Yilmaz & Gabor



Overview

A gaze-controlled virtual keyboard was developed to enable hands-free typing using only a webcam. The system uses eye tracking to determine where a user is looking and blink detection to simulate clicks. This project focuses on improving accessibility and exploring alternative human-computer interaction methods. Traditional input devices such as keyboards and mice can be limiting for users with motor impairments, so this system provides a more inclusive way to interact with digital interfaces.

Methods

The system combines multiple technologies to create a real-time interaction pipeline. WebGazer is used to estimate gaze position from webcam input. MediaPipe FaceMesh is used to detect facial landmarks and calculate blink events. A predictive text system powered by the Datamuse API suggests words based on user input. The gaze data is processed through multiple filtering steps to improve stability and accuracy. These components are integrated using HTML, CSS, and JavaScript to create a fully interactive web-based system.

Results

The final system successfully allows users to type using only gaze and blink input. The gaze tracking became significantly more stable after applying filtering techniques, especially with the addition of the Kalman filter. The predictive text system improved typing speed, and UI adjustments reduced selection errors. While the system is not perfectly precise, it demonstrates that low-cost webcam-based eye tracking can be used for functional interaction.

System Architecture

The system is designed as a real-time pipeline that integrates multiple components to enable gaze-based interaction. Webcam input is first processed by WebGazer to estimate the user's gaze position on the screen. This raw gaze data is then passed through a filtering pipeline consisting of median filtering, speed-based outlier rejection, adaptive smoothing, and a Kalman filter to improve stability and accuracy. The filtered gaze coordinates are used to control a visual cursor, which interacts with the user interface. In parallel, MediaPipe FaceMesh processes the same webcam feed to detect facial landmarks and calculate blink events, which are used to simulate click actions at the current gaze position. Additionally, the Datamuse API provides predictive text suggestions based on the user's input, which are integrated into the interface to improve typing efficiency. All components work together within a web-based environment to create a seamless, hands-free typing experience.

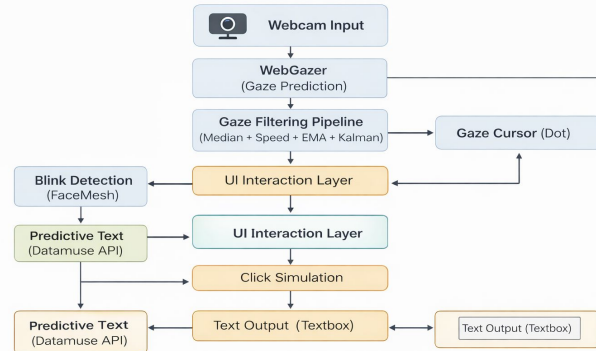


Fig 1. System Architecture of our program

User Interface

The interface was designed to improve usability for gaze-based interaction. The keyboard size was increased to create larger target areas, reducing selection errors. The text box was reduced in size to allow more space for interaction elements. The predictive text buttons were enlarged to make them easier to select. These changes improve both accuracy and comfort when using gaze as an input method.

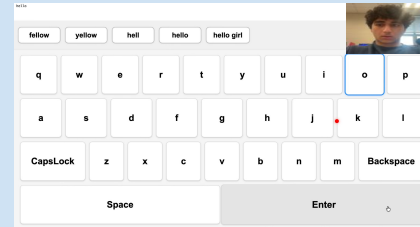


Fig 2. User Interface

Applications

This system could be used in assistive technology for individuals with motor impairments, allowing them to communicate and interact with digital devices using only their eye movements. It could help users type messages, browse the internet, operate software, or control smart devices without relying on traditional keyboards or mice.



Mediapipe